Lecture 0

Hi!

CS 12 Team 25.2
A.Y. 2025–2026, 2nd Sem

Department of Computer Science
College of Engineering
University of the Philippines Diliman

About CS 12

Who are we?

- Jerome Beltran
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  - Lecture
- Daryll Ko
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  - Lecture, Lab

Who are we?

- Miguel Martinez
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  - Lab
- Ren Tristan Dela Cruz
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  - Lab
- Jozelle Addawe
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  - Lab
- Mario Carreon
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  - Lab

CS 12

Course name	CS 12
Course title	Computer Programming II
Units	3.0 (2h lec, 3h lab)
Prerequisite	CS 11 (Computer Programming I)
Prerequisite for	CS 20, CS 32

- Lecture scheds: Tues 9-11 AM (Jem), Thurs 12-2 PM (Daryll)
- Lab scheds:
  - Fri 8:30–11:30 AM (Miguel/Jozelle/Ren)
  - Fri 11:30–2:30 AM (Mario/Daryll)

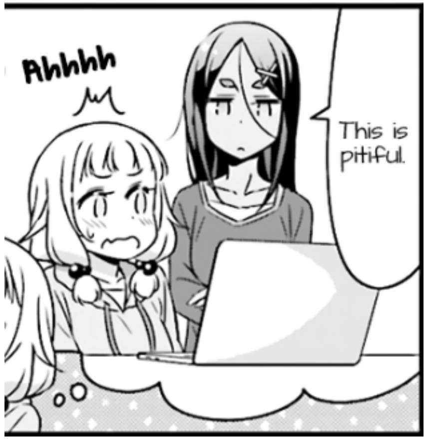
Course Goals

At the end of the course, the student is expected to be able to:

- Design **maintainable software** using abstraction techniques and patterns
- Identify use cases and tradeoffs of **program design practices and paradigms**
- Create **graphical programs** using APIs, frameworks, and best practices
- Develop a sense and appreciation of **well-designed code**

In other words...

If we now know how to write *correct* programs...
...how do we write correct *and organized* programs?



→

source: *New Game!*

High-Level Coverage (optimistic)

Fundamentals

- Program design
- Unit testing and debugging
- Abstraction
- Modularization

Programming paradigms

- Object-oriented programming
  - SOLID, MVC

High-Level Coverage (optimistic)

Programming paradigms (cont’d)

- Functional programming
  - Immutability, higher-order functions
  - Algebraic data types
- Concurrent programming
  - Asynchronous programming
  - Multithreaded programming
  - Shared-state and message-passing concurrency

High-Level Coverage (optimistic)

Application-centric programming

- Event-driven game programming
- Web programming
- `malloc`-based programming

Grading System

Assessment

Requirement	%	#	Delivery
LE	40	1-2	f2f, outside class hours
HOPE	30	1-3	f2f, during class hours
Lab	15	7-15	some f2f, some take-home
Project	15	1-2	take-home

Assessment

- LE—Long Exam (written exam, partly essay-type)
- HOPE—Hands-On Programming Exam (lab exam)
  - **not** OJ-based
- Project—programming project

- LE and HOPE are ***summative***
  - again, **summative**—*checks* whether you understand a topic
- Labs are ***formative***
  - again, **formative**—*helps* you understand a topic better

Grading Scheme

Range	Grade
$96 \leq x$	1.00
$92 \leq x < 96$	1.25
$88 \leq x < 92$	1.50
$84 \leq x < 88$	1.75
$80 \leq x < 84$	2.00

Range	Grade
$75 \leq x < 80$	2.25
$70 \leq x < 75$	2.50
$65 \leq x < 70$	2.75
$60 \leq x < 65$	3.00
$x < 60$	5.00

We do **not** give removal examinations in this course.

Extra Points ❤️

- Some items will be marked with hearts.
  - Heart ❤️ points are easy to get.
  - Heartbreak 💔 points will break your heart…
  - Regular 🔴 points are somewhere in between.

Extra Points ❤️

- These are considered **extra points**, and will be weighted as follows:

Grade	**Weight of extra points**
≥ 2.00	100%
≥ 1.75	40%
≥ 1.50	16%
≥ 1.25	6%
≥ 1.00	2%

Extra Points ❤️

- For each entry (grade range, weight) above, your grade will be evaluated. Your final grade will be the best grade among those five.
  - In particular, extra points can be considered the same as regular (non-extra) points up to a grade of 2.00.
- There will always be enough regular points to get a 100%.

Extra Points ❤️

More precisely, suppose

- R is your total raw score in the regular items, and
- E is your total raw score in the extra items.

Extra Points ❤️

Then your grade is the best (i.e., minimum) among the following:

- $\max(2.00, \text{raw-to-final}(R + E))$
- $\max(1.75, \text{raw-to-final}(R + E \cdot 0.40))$
- $\max(1.50, \text{raw-to-final}(R + E \cdot 0.16))$
- $\max(1.25, \text{raw-to-final}(R + E \cdot 0.06))$
- $\max(1.00, \text{raw-to-final}(R + E \cdot 0.02))$

where "$\text{raw-to-final}(x)$" is the conversion from raw score (x) to final grade (1.00, 1.25, etc.) using the grading scheme table.

More Details

Programming Languages

Python (proficiency expected)

- Object-oriented programming
- Event-driven game programming
- Shared-state concurrency
- Asynchronous programming

We will be using Python **with type annotations**

Programming Languages (cont'd)

Elm (no experience assumed)

- Functional programming
- Message-passing concurrency
- Client-side web programming

C (no experience assumed)

- Lower-level programming
- Multithreaded programming

Platforms

- Google Classroom
  - important announcements
- Discord
  - general communication, Q&A
- DCS OJ (`https://oj.dcs.upd.edu.ph/`)
  - some labs
  - same accounts as CS 11
- DCS-hosted git
  - some labs, HOPE, Project

Policies

Attendance

- Sign every meeting
- Before dropping deadline:
  - `if lec_absences + lab_absences` $\geq$ `8 then DRP`
- After dropping deadline:
  - Grade of 5.00
- Session-long grace period (no distinction between on-time and late)

Attendance

- Only **medical reasons** or **official competitions** are valid reasons for excused absences
  - Email all members of the team with supporting documents (e.g., medical certificate)
  - For absences due to other reasons, kindly use your unexcused absence allowance/budget

Delivery, technical requirements

- By default, lectures and labs will be live (face-to-face).
- Video lectures may be used in some circumstances.
- We expect that students would have the resources to view or download the recorded video lectures, participate in the live discussion classes, as well as do the machine problems.
  - This means having a computer with Internet connection in which you can install stuff.

Misc.

- You will **all** be assessed the same regardless of your status. Do not expect any consideration or special treatment for whatever reason.
- Deliverables can only be submitted prior to the deadline. (**Strict**!)
- Re-hosting of class material (lectures, problems, videos, etc.) is **not** allowed, considered copyright violation.

Academic Dishonesty

- Academic dishonesty will merit a grade of **5.00**
- This includes the use of ChatGPT and similar tools
- An administrative case will be filed against all uncooperative parties
- Handled by the Student Disciplinary Council

Reminder: Honor before excellence

TODOs after this session

- Join Google Classroom
  - https://classroom.google.com/c/NzkzMjYxMDI4Mzgx?cjc=yp6ijmx6
- Join Discord
  - https://discord.gg/EqcdDUH4

TODOs after this session

- Submit **Student Information Form**
  - https://forms.gle/isA3v2A1XzuF5Y177
- Asks for:
  - Basic information
  - Programming experience
  - Medical/Mental health concerns (if you're willing to share)
- Starting **February 1**, attendance will only count if:
  - you have submitted the form, **and**
  - you have opened a ticket in Discord and asked a question

This serves as our syllabus.